

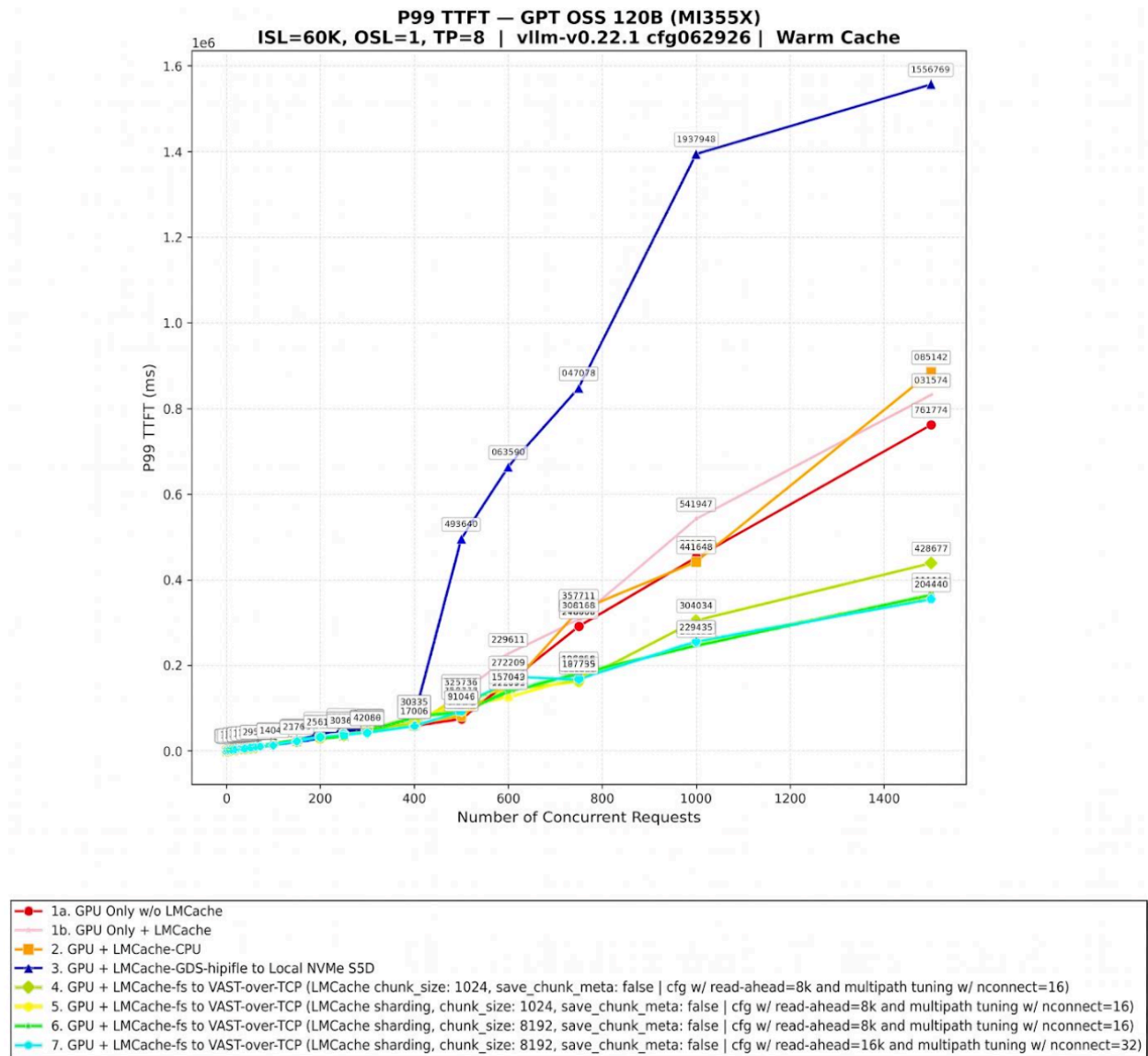
Vast <> Imcache collab

1. We observed performance degradation in GPU-only KV cache configurations when running with LMCache vs vllm only - across both Nvidia and AMD platforms:

- a. NVidia setup:

<https://www.vastdata.com/blog/vastdata-lab-lup-kv-cache-offload-benchmark>

- b. AMD setup:



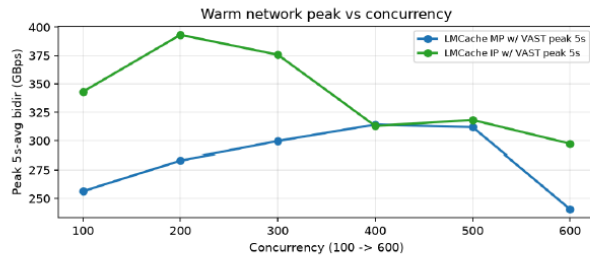
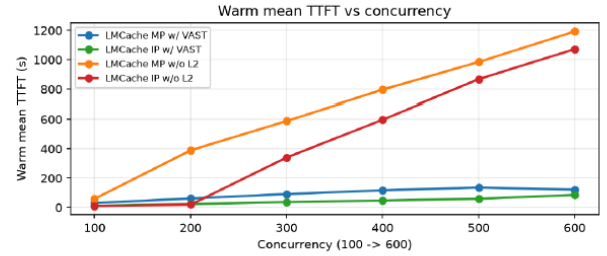
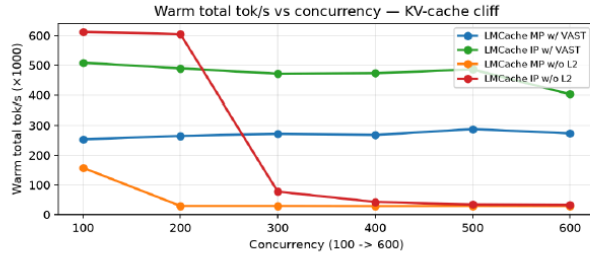
Although this does not impact VAST KV cache offloading, we are sharing these findings for your visibility

2. Performance comparison: LMCache-MP vs. LMCache-IP (performance degradation observed). Tested on AMD GPUs

LMCache ISL 120k · full 4-way matrix (IP + MP)

GPT-OSS-120B TP8 | OSL=1 | hb=60 | drain/gate on | dual-NIC bidir (ens201+ens203) averaged in 5s bins

Charts



Configurations Used

1. IP

None

```
#!/usr/bin/env bash
set -euo pipefail
exec vllm serve /models/model \
  --host 0.0.0.0 --port 8000 \
  --served-model-name gpt-oss-120b \
  --tensor-parallel-size 8 \
  --max-model-len 131072 \
  --enable-prefix-caching \
  --block-size=64 \
  --max-num-seqs 256 \
  --kv-transfer-config \
  '{"kv_connector": "LMCacheConnectorV1", "kv_role": "kv_both"}'
```

None

```
chunk_size: 8192
save_decode_cache: false
enable_async_loading: true
blocking_timeout_secs: 120
lookup_timeout_ms: 240000
local_cpu: false
max_local_cpu_size: 180.0
save_unfull_chunk: false
remote_storage_plugins:
  - fs
store_location: RemoteBackend-fs
extra_config:
  remote_storage_plugin.fs.base_path: /kvcache
  remote_storage_plugin.fs.use_odirect: true
  remote_storage_plugin.fs.read_ahead_size: 8192
  fs_connector_read_ahead_size: 8192
  fs_connector_use_odirect: true
  save_chunk_meta: false
```

2. MP

None

```
lmcache server \
  --host 127.0.0.1 --port 5555 \
  --l1-size-gb 1600 \
  --eviction-policy noop \
  --eviction-trigger-watermark 0.8 --eviction-ratio 0.2 \
  --chunk-size 8192 \
  --l2-prefetch-max-in-flight 4 \
  --max-gpu-workers 8 --max-cpu-workers 8 \
  --worker-reap-timeout-seconds 180 \
  --l2-adapter
'{"type":"fs_native","base_path":"/kvcache","num_workers":4,"use_odirect":true,"max_capacity_gb":0,"read_ahead_size":8192}' \
  --l1-align-bytes 1048576 \
  --l2-store-policy skip_l1 &
```

None

```
exec vllm serve /models/model \
  --host 0.0.0.0 --port 8000 \
  --served-model-name gpt-oss-120b \
  --tensor-parallel-size 8 \
  --max-model-len 131072 \
  --disable-hybrid-kv-cache-manager \
  --enable-prefix-caching \
  --block-size=64 \
  --max-num-seqs 256 \
  --kv-transfer-config \

'{"kv_connector":"LMCacheMPConnector","kv_connector_module_path":"lmcache.integration.vllm.lmcache_mp_connector","kv_role":"kv_both","kv_connector_extra_config":{"lmcache.mp.host":"tcp://127.0.0.1","lmcache.mp.port":5555,"lmcache.mp.heartbeat_interval":60.0}}'
```

None

```
chunk_size: 8192
save_decode_cache: false
enable_async_loading: true
blocking_timeout_secs: 120
lookup_timeout_ms: 240000
mp_host: 127.0.0.1
mp_port: 5555
```

4. Prometheus logging : We got feedback from a customer that would appreciate more logging and metrics observability through prometheus on lmcache